

AVICHAL BAWEJA

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SUMMARY

Computer Vision Engineer with strong expertise in developing drone vision systems, OCR pipelines, and real-time image processing applications. Proficient in YOLOv5, OpenCV, TensorFlow, and computer vision frameworks. Experienced in building AI-powered solutions, deploying CV models in production environments, and developing end-to-end image processing pipelines for enterprise applications.

TECHNICAL SKILLS

Computer Vision: OpenCV, YOLOv5, TensorFlow, PyTorch, MediaPipe, Image Processing, Object Detection, Segmentation

OCR/OMR: Tesseract, EasyOCR, Custom OCR Models, Document Analysis, Exam Processing, Data Extraction

Programming: Python, C++, C, MATLAB, CUDA, Arduino, SQL, Bash

Tools & Platforms: ROS, Gazebo, Docker, Git, Linux, FastAPI, GCP, AWS, Vector Databases, CI/CD

EXPERIENCE

Founding Engineer | *Datavedam*

Oct 2025 – Present

- Built OCR and OMR pipelines for automated document scanning and exam processing in an AI-driven school ERP platform with quality validation and automated workflows.
- Implemented QR-code based admission workflow system for automated processing, verification, and enrollment of student documents at scale.
- Developed AI agents for automated administrative tasks, intelligent assistance, operational automation, and real-time contextual responses.
- Built RAG pipeline converting CBSE circulars into vector embeddings for intelligent querying, semantic search, and decision support.
- Designed scalable backend services for computer vision applications using Python and FastAPI with microservices architecture.

Business Analyst | *Purshotam Profiles Pvt. Ltd.*

Jan 2025 – July 2025

- Identified and evaluated new business opportunities in the solar projects sector; prepared proposals, presentations, and performance reports for client engagement.
- Collaborated with cross-functional teams to optimize sales processes, improve operational efficiency, and strengthen B2B client relationships.

Computer Vision Intern | *TechEagle Innovations Pvt. Ltd.*

Apr 2024 – May 2024

- Developed a computer vision-based drone system integrating YOLOv5 with real-time feedback control for stability, autonomous navigation, and obstacle avoidance.
- Designed and simulated an autonomous landing system in ROS and Gazebo using Python, OpenCV, and ArUco markers for precise coordinate-based landings.

IT Intern | *Sankalptaru Foundation*

Jan 2023 – Mar 2023

- Built predictive models using Python, Pandas, and scikit-learn for forecasting farm conditions; performed statistical demand analysis with Matplotlib to support agroforestry.

PROJECTS

Drone Computer Vision System | *YOLOv5, OpenCV, ROS, Gazebo, Python*

- Integrated YOLOv5 with real-time feedback control for drone stability, autonomous navigation, and obstacle avoidance in simulation environments.
- Designed and simulated autonomous landing using ArUco markers with coordinate-based precision landing achieving sub-meter accuracy.

Datavedam – AI Powered School ERP | *Python, FastAPI, OCR, AI Agents*

- Built OCR/OMR pipelines for document and exam processing with automated quality validation and intelligent data extraction workflows.
- Developed AI agents for automated administrative tasks; built RAG pipeline for CBSE circulars with vector embeddings.

Handwritten Text to LaTeX Converter | *Python, OCR, LLM, Image Processing*

- Built OCR + LLM pipeline to extract handwritten notes and generate accurate LaTeX documents with optimized prompts for math expressions.

Trackpad Control using Hand Gestures | *Python, OpenCV, MediaPipe, PyAutoGUI*

- Developed real-time hand gesture-based trackpad control system with efficient hand tracking, high-accuracy gesture detection, and smooth interaction.

EDUCATION

Christ University

B.Tech. Computer Science & Engineering, Honors in AI & ML

Bengaluru, Karnataka

2021 – 2025

Delhi Public School

Senior Secondary School (CBSE)

Agra, Uttar Pradesh

2019 – 2021